

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

4 Complexity Theory (ad260)

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a *constructible* function.

Nondeterminism.
Space Complexity

- (a) Explain why $\text{NTIME}(f) \subseteq \text{SPACE}(f)$ [4 marks]

Answer: The statement means that any language that can be recognized by a non-deterministic machine in time $O(f)$ can also be recognized by a deterministic machine using space $O(f)$. A deterministic machine can simulate a nondeterministic machine M by backtracking. In order to do this, it has to keep track of the current configuration of M as well as all the choices taken wherever a nondeterministic choice was available to M . The space required to record these choices is clearly no more than a constant multiple of the length of the computation. Moreover, the space required to store the configuration of M also cannot be more than the number of steps in the computation of M so far. So, the total work space required by the simulating machine is $O(f)$, where f is the time bound on M .

Nondeterminism.
Reachability
Method.

- (b) Explain why any language $L \in \text{NSPACE}(f)$ is also in $\text{TIME}(c^{(f(n)+\log n)})$ for some constant c . [6 marks]

Answer: Suppose M is a nondeterministic machine working with workspace bounded by $f(n)$ for inputs of length n . For a given input string x of length n , the total number of distinct configurations of M is no more than $qn f(n) s^{f(n)}$, where s is the size of the tape alphabet. For some constant d , this is less than $nd^{f(n)}$. The configuration graph of M on input x has these configurations as nodes, and edges corresponding to one-step transitions. We can simulate M deterministically by solving the reachability problem on this graph. This can be done in time $O(g^2)$, where g is the size of the configuration graph. That is at most $d'(nd^{f(n)})^2$ for some constant d' . But this is $c^{\log n + f(n)}$, for some constant $c \leq d'd^2$.

- (c) A deterministic machine is a special case of a non-deterministic machine. Explain what this and the results above tell us about the inclusions among the following complexity classes:

$L, NL, P, NP, PSPACE, NPSPACE$.

[5 marks]

Answer: We can establish the following sequence of inclusions

$$L \subseteq NL \subseteq P \subseteq NP \subseteq PSPACE \subseteq NPSPACE.$$

Of these, the first, third and fifth follow from the definitions, since a deterministic machine is a special case of a non-deterministic machine.

The second follows from part (b) above since if $f = O(\log n)$, then $c^{(f(n)+\log n)}$ is bounded by a polynomial in n .

The fourth follows from part (a).

- (d) Savitch proved that there is an algorithm for the graph reachability problem that uses $O((\log n)^2)$ space. What further inclusions can you derive among the above complexity classes using this fact? Explain your answer. [5 marks]

Answer: We can use Savitch's algorithm to show that $\text{NSPACE}(f) \subseteq \text{SPACE}(f^2)$ for any constructible f with $f(n) = \Omega(\log n)$. This is because we can use the algorithm to solve the reachability problem on the configuration graph of a nondeterministic machine M on input x using space $O((\log g)^2)$ where g is the number of configurations. A calculation similar to that in part (b) shows that this is $O(f^2)$ when M uses space $O(f)$. We have to be careful in doing this not to construct the entire configuration graph, but just check for individual configurations whether there is a valid transition between them.

This, in particular, establishes that $\text{NPSpace} \subseteq \text{PSpace}$, and thus that $\text{PSpace} = \text{NPSpace}$.
